

Bulk Ultrasonic Water Meter (DN50-400)

MODBUS Communication Protocol

V1.23

The ultrasonic water meter adopts the MODBUS-RTU communication protocol. The MODBUS protocol specifies in detail the checksum and data sequence, which are essential components for specific data exchange. The MODBUS protocol uses a master-slave responder connection (half-duplex) over a single communication line, meaning signals are transmitted in opposite directions on the same communication line. First, the signal from the host computer is sent to a unique terminal device (slave), and then the response signal from the terminal device is transmitted back to the host in the opposite direction.

The MODBUS protocol only allows communication between the host (computer, PLC, etc.) and the terminal devices, and does not permit direct data exchange between independent terminal devices. This ensures that terminal devices do not occupy the communication line during initialization and only respond to query signals addressed to them.

The MODBUS-RTU format does not include start or end characters but requires the addition of a waiting time. The waiting time must be at least 3.5 characters.

Precautions:

1. The meter requires DC for communication and power supply, and the voltage must be maintained between 12V and 24V. If the voltage exceeds the permissible limit, the meter may be damaged.
2. The frequency of meter reading should not be too high to avoid affecting the measurement accuracy of the meter and the lifespan of the battery.

Transmission Method

Information is transmitted asynchronously in bytes.

The communication messages exchanged between the master and slave devices follow a 10-bit format, which includes:

- 1 start bit, 8 data bits (the least significant bit is sent first), no parity bit, and 1 stop bit. The default baud rate is 9600 bps.

Data frame format

Address	XX	1byte
Function Code	0x03	1byte
Starting Register High Address	0x10	n bytes
Starting Register Low Address	0x10	n bytes
Number of Registers High	0x00	n bytes
Number of Registers Low	0x0C	n bytes
CRC Check Low Byte	XX	2 bytes
CRC Check High Byte	XX	

Address:

The meter address can be configured within the range of 1 to 247 (Note: Address 104 is reserved and cannot be used). In hexadecimal representation, this range is 0x01 to 0xF7. These bits represent the user-assigned address of the terminal device that will receive data from the connected host. Each terminal device must have a unique address, and only the addressed terminal will respond to a query containing its address. When the terminal sends a response, the slave address data in the response informs the host which terminal it is communicating with.

Function Code	Definition	Control Function
03H	Read Data Registers	Retrieve the current binary value of one or more registers

Function Code:

The function code informs the addressed terminal (slave) about the specific operation to be performed. The table below lists the function codes used in this series of meters, along with their meanings and functions.

Data Field:

The data field contains the information required for the terminal to execute the specified function or the data collected by the terminal in response to a query. The content of this field may include values, reference addresses, or setting parameters. For example, when the function code instructs the terminal to read registers, the data field specifies the starting register address and the number of registers to read. The embedded address and data may vary depending on the type of slave device and its specific content.

CRC Checksum:

The CRC (Cyclic Redundancy Check) checksum is calculated for the data frame, and the resulting two-byte checksum is appended to the end of the frame. The low byte of the CRC is placed first, followed by the high byte. The CRC calculation covers all data from the "Address" field to the "Low Byte of Register Quantity." The CRC field occupies 2 bytes and contains a 16-bit binary value, with the low 8 bits transmitted first and the high 8 bits transmitted last. The CRC value is calculated by the transmitting device and appended to the data frame. The receiving device recalculates the CRC during data reception and compares it with the received CRC value. If the two values do not match, an error is indicated. The CRC calculation range includes all data from the address field to the data preceding the CRC field.

In summary, for a Modbus data frame sent from the master, only two parts are variable: the address field and the CRC checksum.

CRC Generation Process:

Initialize a 16-bit register with 0xFFFF (all bits set to 1), referred to as the CRC register.
Perform an XOR operation between the 8-bit data of the first byte in the frame and the low byte of the CRC register, then store the result back into the CRC register.
Shift the CRC register one bit to the right, filling the highest bit with 0 and checking the shifted-out lowest bit.

If the lowest bit is 0, repeat step 3 (perform the next shift). If the lowest bit is 1, perform an XOR operation between the CRC register and a fixed preset value (0xA001).

Repeat steps 3 and 4 until 8 shifts are completed. This completes the processing of one byte.

Repeat steps 2 to 5 to process the next byte until all bytes in the frame are processed.

The final value in the CRC register is the CRC checksum.

Additionally, there is a method for calculating CRC using a predefined lookup table, which is faster but requires more storage space. This method is not discussed here; please refer to relevant documentation for details.

(3) Communication Request Format

Function Code 03H: Read Registers

This function code allows the user to retrieve data collected and recorded by the device, as well as system parameters. The host can request any amount of data in a single request, but the requested data must not exceed the defined address range.

The following example demonstrates reading basic data from a slave device with address 01 (each address in the data frame occupies 2 bytes). The collected data is the forward cumulative flow (occupying 2 registers, totaling 4 bytes), with register addresses 4118 and 4119. The corresponding hexadecimal value for these addresses is 0x1016.

The data frame for the issued command is as follows:

01 03 10 16 00 02 21 0F

Host sending		Send a message
Address code		01H
Function code		03H
Start address	High byte	10H
	Low byte	16H
Number of registers	High byte	00H
	Low byte	02H
CRC check code	Low byte	21H
	High byte	0FH

Slave return		Return information
Address code		01H
Function code		03H
Number of bytes		04H
Register data	High byte	00H
	Low byte	12H
Register data	High byte	D6H
	Low byte	87H
CRC Check code	Low byte	44H

	High byte	34H
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Explanation of Data Representation and Format:

In the table, the measured reading is represented by 4 bytes and is expressed in cubic meters (m³). The data follows the "high byte first, low byte later" convention. For example, if the read value is **0012D687H** (1234567 in decimal), the measured value is calculated as: 1234567 × 0.01 = 12345.67 m³.

The meter collects data based on the data frame sent by the master station and constructs a Modbus data frame to return to the master. The meter fills the "data field" in the table, ensuring that each data item follows the high-byte-first, low-byte-later rule. Below are examples to illustrate this:

Example 1:

Data Item (Decimal):12345

Hexadecimal Representation: 0x3039Byte Representation (2 bytes):

High byte: 0x30 Low byte: 0x39

Byte Order in Data Frame: 0x30 0x39

Example 2:

Data Item (Decimal): 1234567

Hexadecimal Representation: 0x12D687

Byte Representation (4 bytes):

o High byte of the high word: 0x00

Low byte of the high word: 0x12

High byte of the low word: 0xD6。

Low byte of the low word: 0x87D

Byte Order in Data Frame: 0x00 0x12 0xD6 0x87

(4)MODBUS register allocation list

The instrument has 15 register addresses within the range of 4112 to 4126 (represented in hexadecimal as 0x1010 to 0x101B). Each register has a data length of two bytes.**If tested with modscan software, the register address bit needs to be read starting from 4113.**

Register address Hex	Register address Dec	Register count	Variable name	Unit	Number of bytes
1010H	4112	1	Battery voltage	0.01V	2 bytes
1011H	4113	1	Water supply temperature	0.01℃	2 bytes
1012H 1013H	4114-4115	2	Reverse cumulative flow	L	4 bytes
1014H 1015H	4116-4117	2	Reverse instantaneous flow	L	4 bytes
1016H 1017H	4118-4119	2	Forward cumulative flow	L	4 bytes

1018H 1019H	4120-4121	2	Forward instantaneous flow	L	4 bytes
1020H	4122	1	Alarm status	High byte ST1	2 bytes
1021H	4123	1	Alarm status	Low byte ST1	2 bytes

(5) Table of units

Cumulative flow unit:

0C 10	1mL
0C 11	10mL
0C 12	100mL
0C 13	1L

Instantaneous flow unit:

0B 38	1mL/h
0B 39	10mL/h
0B 3A	100mL/h
0B 3B	1L/h

Description of the data item filled in the data area:

Data item	Unit	Caliber	Decimal point	Number of bytes	Group the magnification of the data frame
Water temperature	°C	All	0.01	2	100
Cumulative flow	m ³	DN50-400	0.001	4	10
Instantaneous flow	m ³ /h	DN50-150	0.01	4	100
		DN200-400	0.1	4	10

(6)Data frame example

When the meter address (S/N) is 1:

Host sends data frame: 01 03 10 10 00 0C 40 CA

Meter returns data frame:

01 03 18 01 68 12 02 00 BC 61 4E 00 CB 21 34 00 45 9B 7D 03 62 88 14 00 20
03 00 67 B5

Meter returns data frame	Instructions (Hexadecimal to decimal)
01	Mailing address
03	Function code
18	Byte length (register)
01 68	Battery voltage (3.57V)
12 02	Water temperature (46.1°C)
00 BC 61 4E	Reverse cumulative flow (4561.789m ³)
00 CB 21 34	Reverse instantaneous flow (14.356m ³ /h)
00 45 9B 7D	Forward cumulative flow (4561.789m ³)

00 62 88 14	Forward instantaneous flow (14.356m³/h)
00 20	Alarm status ST1 (High byte reserved, 00 not parsed)
03 00	Alarm status ST2 (Low byte reserved, 00 not parsed)
67 B5	CRC check

(7)Status word parsing

ST1

Bit 0	0 normal 1 Flow overload	Bit 4	Reserve
Bit 1	Reserve	Bit 5	0 normal 1 Lack of water
Bit 2	0 normal 1 undervoltage	Bit 6	Reserve
Bit 3	Reserve	Bit 7	Reserve

ST2

Bit 0	0 normal 1 Sensor ① lack of water	Bit 4	Reserve
Bit 1	0 normal 1 Sensor ② lack of water	Bit 5	Reserve
Bit 2	Reserve	Bit 6	Reserve
Bit 3	Reserve	Bit 7	Reserve

Parsing example:

Register 4122 has a value of 0020 ST1=20H

20H to BIN= 0010 0000 (From right to left, Bit0,Bit1,...Bit7)

Parsing by status word:

Bit5=1 Lack of water

Register 4123 has a value of 0003 ST2=03H

03H to BIN= 0011 (From right to left, Bit0,Bit1,...Bit7)

Parsing by status word:

Bit0=1 Sensor ① lack of water Bit1=1 Sensor ② lack of water

Note: If ModScan is used for communication testing, the starting address of the registers should be observed beginning from 4112.

Wiring Method: Power supply standard:DC 12-24V



The communication address of the instrument is derived by taking the last two digits of the S/N (Serial Number). The communication address is displayed in hexadecimal format. If necessary, please convert it to decimal format, referring to the data sheet below.

Note: In the S/N, 68 corresponds to 104, and 00 corresponds to 0. These two addresses are unavailable for use.

Communication Address Mapping Table

HEX (S/N)	DEC	HEX (S/N)	DEC	HEX (S/N)	DEC
01	1	34	52	67	103
02	2	35	53	68	104
03	3	36	54	69	105
04	4	37	55	70	112
05	5	38	56	71	113
06	6	39	57	72	114
07	7	40	64	73	115
08	8	41	65	74	116
09	9	42	66	75	117
10	16	43	67	76	118
11	17	44	68	77	119
12	18	45	69	78	120
13	19	46	70	79	121
14	20	47	71	80	128
15	21	48	72	81	129
16	22	49	73	82	130
17	23	50	80	83	131
18	24	51	81	84	132
19	25	52	82	85	133
20	32	53	83	86	134
21	33	54	84	87	135
22	34	55	85	88	136
23	35	56	86	89	137
24	36	57	87	90	144
25	37	58	88	91	145
26	38	59	89	92	146
27	39	60	96	93	147
28	40	61	97	94	148
29	41	62	98	95	149
30	48	63	99	96	150
31	49	64	100	97	151
32	50	65	101	98	152

33	51	66	102	99	153
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